When F1 Fails: Granularity-Aware Evaluation for Dialogue Topic Segmentation

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Abstract

Dialogue topic segmentation supports summarization, retrieval, memory management, and conversational continuity. Despite decades of prior work, evaluation practice in dialogue topic segmentation remains dominated by strict boundary matching and F1-based metrics [1, 2], even as modern LLM-based conversational systems increasingly rely on segmentation to manage conversation history beyond the model's fixed context window, where unstructured context accumulation degrades efficiency and coherence.

This paper introduces an evaluation objective for dialogue topic segmentation that treats boundary density and segment coherence as primary evaluation criteria, used alongside tolerance-based boundary accuracy. Through extensive cross-dataset empirical evaluation, we show that apparent performance differences across dialogue segmentation benchmarks are driven not by model quality, but by mismatches in annotation granularity and sparsely annotated topic boundaries.

We evaluated multiple, structurally distinct dialogue segmentation strategies across eight dialogue datasets spanning task-oriented, open-domain, meeting-style, and synthetic interactions. Across these settings, we observe high segment coherence combined with extreme oversegmentation relative to sparse labels, producing misleadingly low exact-match F1 scores. We show that topic segmentation is best understood as selecting an appropriate granularity rather than predicting a single correct boundary set. We also propose a topic segmentation algorithm that separates boundary scoring from boundary selection, and demonstrate that this separation makes boundary density a controllable design choice and clarifies why thresholds fail to transfer across datasets.

1 Introduction

Dialogue topic segmentation is commonly framed as identifying points in a conversation where "the subject changes." This framing is appealingly simple. It is also incorrect. Topic structure is often gradual, nested, and perspective-dependent.

Despite this, many dialogue segmentation benchmarks treat annotated boundaries as objective ground truth and evaluate systems using strict boundary matching and F1-based metrics [1]. These metrics collapse distinct failure modes into a single number. Small boundary shifts are often harmless when the goal is coherent topic blocks, while missing or adding a few boundaries can be catastrophic if boundaries drive downstream actions such as summarizing or dropping earlier turns.

This paper addresses that mismatch by changing what we report and what we optimize. We define an evaluation objective that jointly reports (i) tolerant boundary accuracy, (ii) boundary density, and (iii) segment coherence. We also propose an algorithmic separation between boundary scoring and boundary selection that makes boundary density explicit and controllable. Together, these contributions let us distinguish detection failures from granularity mismatches and explain why a single global threshold fails to transfer across datasets.

We emphasize, however, that boundary-based metrics such as F1 are appropriate when annotation granularity is consistent between training, evaluation, and intended use. The failure modes illustrated here arise when segmentation granularity differs across datasets or between system behavior and annotation intent, which is increasingly common in dialogue benchmarks and downstream applications.

1.1 Motivation: Conversational Memory in AI Chat Systems

Large language model (LLM) based chat systems operate under finite context windows, which require practical systems to manage conversational history explicitly. As conversations grow longer, earlier turns must be dropped, summarized, or selectively retrieved. Topic segmentation decisions therefore directly determine which portions of prior dialogue are preserved, compressed, or reintroduced.

Importantly, this challenge does not disappear as context windows grow larger. Unstructured accumulation of prior turns increases computational cost, dilutes relevant context, and degrades the model's ability to focus on salient information. As a result, effective conversational systems must impose structure on dialogue history rather than relying on raw context expansion.

In this setting, topic segmentation functions as a memory control mechanism rather than a purely linguistic task. Poor segmentation can cause relevant context to be discarded prematurely or irrelevant context to be retained, leading to degraded long-horizon coherence and unstable behavior across turns. The absence of robust episodic memory remains a major limitation of current LLM-based systems, particularly in multi-session interactions.

This work is motivated by the development of *Episodic*, a conversational memory system that organizes dialogue into topic-structured representations to support persistent interactions with LLM-based assistants [3]. Within such systems, boundary placement governs summarization, retrieval, and context reintegration, making segmentation granularity a first-order systems design decision rather than an evaluation detail.

2 Background and Related Work

Topic segmentation has a long history in text processing, with early approaches such as TextTiling [4] and Choi-style concatenation benchmarks [5]. Boundary-based evaluation measures such as Pk [6] and WindowDiff [7] formalized strict alignment metrics.

Dialogue topic segmentation introduces additional complexity due to turn-taking structure and discourse phenomena. Dialogue benchmarks include SuperDialseg [8], DialSeg711 [9], and TIAGE [10], each reflecting distinct annotation philosophies and boundary densities. Nevertheless, evaluation practice in dialogue segmentation remains centered on precision, recall, and exact-match F1 computed over boundary positions [1].

3 Problem Setting

Topic boundaries are not intrinsic properties of conversations. Instead, they depend on segmentation granularity, task requirements, and annotation intent. A model that produces fine-grained discourse-phase boundaries can be operationally reasonable yet score poorly against a dataset that annotates only coarse task transitions. Conversely, a model tuned to sparse labels can suppress meaningful structure.

This paper treats topic segmentation as two separate operations:

1. **Boundary scoring:** assign a score to each candidate boundary position.

2. **Boundary selection:** choose which candidates become output boundaries, controlling boundary density and spacing.

Collapsing these operations into a single evaluation target makes boundary density an accidental side effect of threshold tuning and yields systematically misleading conclusions under annotation mismatch.

4 Datasets

We evaluated multiple, structurally distinct dialogue segmentation strategies across eight dialogue datasets spanning diverse dialogue regimes. The purpose of cross-dataset evaluation here is not only to compare performance, but to expose how annotation philosophy and boundary density interact with evaluation metrics.

Task-oriented dialogue datasets include SuperDialseg [8], DialSeg711 [9], TIAGE [10], and MultiWOZ [11]. Open-domain and semi-structured datasets include DailyDialog (synthetic concatenations) [12], Taskmaster [13], Topical-Chat [14], and QMSum [15]. Several datasets provide sparse boundaries by design or annotate boundaries with intent (e.g., summarization units), which is precisely the setting where exact-match F1 becomes fragile.

5 Methodology

Our approach outputs a score for each canonical between-message position indicating the likelihood of a topic boundary. We map all gold and predicted boundaries to canonical between-message indices to avoid speaker-based off-by-one artifacts.

The key methodological choice is to separate boundary scoring from boundary selection. Scoring aims to generalize: it should rank likely topic transitions above unlikely ones. Selection is a decision rule: it chooses boundary density and spacing to match an annotation regime or an application requirement.

5.1 Topic Segmentation Algorithm

Given a dialogue consisting of messages $m_1, \dots, m_T$, define candidate positions $i \in \{1, \dots, T-1\}$ between adjacent messages.

Boundary scoring. Compute scores $s_i \in \mathbb{R}$ for all candidate positions. Larger s_i indicates stronger evidence of a topic boundary.

Boundary selection (rule used in experiments). We use thresholding with minimum spacing:

1. Select all candidates i with $s_i \geq \tau$.

2. Enforce minimum spacing g by greedy non-maximum suppression: process selected candidates in descending score order and accept i only if no previously accepted boundary lies within g positions.

The output boundary set is $B \subseteq \{1, \dots, T-1\}$.

For Figure 2, we use the same selection rule but adapt the threshold over time; the adaptive threshold is logged as *min_evidence*.

5.2 Scoring Model Architecture

The boundary scoring model is a fine-tuned DistilBERT-based binary classifier. For each candidate boundary position i, the model encodes the adjacent message pair (m_i, m_{i+1}) using the input format `[CLS]` m_i `[SEP]` m_{i+1} `[SEP]`. The final hidden state of the `[CLS]` token is passed through a linear projection layer to produce a scalar score $s_i \in \mathbb{R}$ representing boundary confidence.

The model is trained using binary cross-entropy loss on boundary labels. No architectural modifications beyond the classification head are introduced; the purpose of this implementation is to provide a stable scoring signal rather than to optimize segmentation performance.

5.3 Training Procedure

We train the scoring model in three stages:

Stage 1: Synthetic Pretraining. We construct synthetic splice examples by concatenating real dialogue segments and labeling the splice point as a boundary. We remove boundary-local artifacts (e.g., greetings) to prevent spurious cues, and we evaluate a negative control set of single-segment examples to verify that the model does not hallucinate boundaries.

Stage 2: Supervised Fine-Tuning on Benchmarks. We next evaluate the effect of supervised fine-tuning on benchmark datasets. Table 3 summarizes performance over five epochs.

Stage 3: Temperature Calibration. We apply temperature scaling to calibrate score magnitudes by optimizing a single scalar temperature $T > 0$ on a held-out validation set (minimizing negative log-likelihood). This operation rescales the logits used to produce boundary scores, improving score comparability and numerical stability across runs.

Importantly, temperature calibration rescales *evidence* but does not determine how many boundaries are produced. Boundary density is controlled exclusively by the selection rule (thresholding and spacing), not by temperature. While changing T shifts the score distribution, it does not encode a target boundary rate and cannot, by itself, enforce consistent segmentation granularity across datasets. Calibration therefore improves the interpretability of scores but does not resolve granularity mismatch, which remains a decision-level issue handled by explicit boundary selection.

Reproducibility. The reference implementation for this work is available in the `paper/` directory of the Episodic repository at `https://github.com/mhcoen/episodic`. This directory contains the LaTeX source for the paper, all experiment configuration files, evaluation scripts, and code required to reproduce the reported results.

Data availability. All datasets used in this study are publicly available from their original sources and are cited in the paper. Due to licensing restrictions, we do not redistribute raw dialogue data. Instead, the repository provides scripts to download, preprocess, and evaluate each dataset in accordance with its original distribution terms.

6 Evaluation Metrics

Relation to Pk and WindowDiff. Classic segmentation metrics such as Pk and WindowDiff [6, 7] were designed to measure boundary placement accuracy under a single assumed segmentation granularity, and they remain useful for diagnosing alignment errors (e.g., off-by-one boundary shifts) in homogeneous annotation settings. However, like exact-match F1, these metrics implicitly conflate detection quality with granularity alignment: they penalize segmentations that produce additional boundaries even when those boundaries correspond to coherent subtopics omitted by a coarser gold standard.

The evaluation objective proposed here is not a drop-in replacement for Pk or WindowDiff. Instead, it decomposes segmentation behavior into complementary dimensions: tolerant boundary detection (W-F1), boundary density relative to the annotation scheme (BOR), and segment-level coherence (purity and coverage). This decomposition makes it possible to distinguish genuine detection failures from granularity mismatch, a distinction that alignment-based metrics alone cannot express. In practice, Pk or WindowDiff may still be reported for continuity with prior work, but we argue that they should be interpreted alongside explicit boundary density and coherence diagnostics when annotation granularity varies across datasets or applications.

We report three evaluation measures:

- **W-F1:** window-tolerant F1 [7], matching predicted boundaries to gold boundaries within a fixed window.
- **BOR:** boundary oversegmentation ratio, $\mathrm{BOR} = \#B_{\mathrm{pred}}/\#B_{\mathrm{gold}}$.
- **Purity/Coverage:** segment coherence diagnostics from clustering evaluation.

Unless otherwise stated, W-F1 is computed using a symmetric tolerance window of ± 1 message around each gold boundary, following common practice in dialogue segmentation to account for minor annotation and indexing variation. We found this window size to balance robustness to off-by-one annotation differences without masking systematic oversegmentation effects, which are captured separately by BOR.

Segment-level coherence computation. To apply purity and coverage to dialogue segmentation, we treat both predicted and gold segmentations as sequences of contiguous spans between boundaries. Purity is computed by assigning each predicted segment to the gold segment with which it has maximum overlap and measuring the fraction of turns in the predicted segment belonging to that gold segment. Coverage is computed symmetrically by measuring, for each gold segment, the fraction of its turns captured by a single predicted segment. These measures assess whether segments are internally coherent, independent of exact boundary placement.

Strict exact-match boundary F1 is reported only as a baseline, because it collapses under coarse or sparse annotation and penalizes coherent segmentations at a different granularity. The evaluation objective in this paper treats boundary density (BOR) and segment coherence (purity/coverage) as primary criteria alongside tolerant boundary accuracy (W-F1).

Interpreting metrics through failure modes. W-F1 detects gross misalignment; BOR exposes systematic over- or under-segmentation; purity/coverage indicate whether the resulting segments are coherent and usable. Low F1 with high purity and high BOR typically indicates granularity mismatch rather than incoherent segmentation.

6.1 Baselines

This subsection examines how simple non-semantic baselines behave under the proposed evaluation objective. The goal is not to establish competitive performance, but to demonstrate how boundary density alone can dominate traditional boundary metrics.

By comparing the proposed model against trivial strategies such as predicting no boundaries, fixed-interval boundaries, or random boundaries matched to the gold density, we isolate which aspects of performance are attributable to semantic boundary detection versus density alignment. Table 1 summarizes these results.

Notably, these baselines highlight a fundamental limitation of boundary-based metrics: when boundary density is matched to the gold annotation, even strategies that ignore the underlying distribution of gold boundaries can achieve deceptively strong F1 and W-F1 scores.

- **No-boundary:** predict no boundaries.
- **Every-N:** predict a boundary every N positions.
- **Random-at-rate:** predict boundaries at a fixed rate matched to the gold boundary density.

Baseline	Dataset	W-F1	BOR	Purity	Coverage
No-boundary	DialSeg711	0.000	0.00	0.438	1.000
	SuperSeg	0.113	0.00	0.568	1.000
	TIAGE	0.140	0.00	0.541	1.000
Every-N	DialSeg711 (N=4)	0.639	0.97	0.812	0.823
	SuperSeg (N=2)	0.623	1.10	0.830	0.775
	TIAGE (N=3)	0.610	0.97	0.849	0.774
Random (BOR=1)	DialSeg711	0.528	1.00	0.839	0.833
	SuperSeg	0.678	1.00	0.883	0.884
	TIAGE	0.664	1.00	0.873	0.869

Table 1: Baseline segmentation results. Random and periodic baselines achieve competitive W-F1 and purity when boundary density matches the gold distribution, demonstrating that boundary density alignment alone can produce deceptively strong scores under standard metrics.

Epoch	F1	W-F1	BOR	Salience	Neg. Ctrl.
1	0.265	0.420	0.31	0.007	0.000
2	0.453	0.688	0.67	0.009	0.032
3	0.489	0.733	0.87	0.010	—

Table 2: Stage 1 synthetic pretraining on splice boundaries. "Salience" denotes the mean boundary score magnitude $\mathbb{E}[|s_i|]$ on the validation set and serves as a diagnostic for score collapse. Negative control ("Neg. Ctrl.") reports the predicted boundary rate on single-segment dialogues, which should be near zero; it was not rerun for epoch 3, as the model had already passed this sanity check at epoch 2.

7 Results

This section evaluates the proposed separation between boundary scoring and boundary selection under heterogeneous annotation granularity.

7.1 Stage 1: Synthetic Pretraining on Splice Boundaries

This stage isolates whether the scoring model can learn a meaningful boundary signal in a controlled setting, independent of annotation noise. Performance here reflects the model's ability to distinguish genuine topic transitions from non-boundaries before exposure to real dialogue benchmarks. Table 2 summarizes boundary-level behavior during this phase.

7.2 Stage 2: Supervised Fine-Tuning on Benchmarks

This stage evaluates how supervised fine-tuning aligns the scoring model with human annotation conventions on in-distribution datasets. The key question is whether boundary density stabilizes without degrading boundary ranking quality. Table 3 reports performance over five fine-tuning epochs.

7.3 Stage 3: Final Test with Calibration

This stage evaluates generalization under distribution shift, where annotation granularity differs from the training regime. Here, the focus is not peak F1 but the interaction between boundary ranking, boundary

Epoch	Overall F1	W-F1	BOR	SuperSeg F1/BOR	TIAGE F1/BOR
1	0.778	0.833	1.11	0.788/1.13	0.197/0.24
2	0.819	0.850	1.23	0.830/1.25	0.352/0.86
3	0.840	0.865	1.13	0.852/1.14	0.337/0.82
4	0.848	0.875	1.08	0.862/1.08	0.322/0.78
5	0.843	0.872	1.04	0.857/1.05	0.311/0.84

Table 3: Stage 2 supervised fine-tuning on benchmark datasets. Boundary density converges toward annotated density (BOR ≈ 1) on datasets seen during training, while W-F1 remains high.

Dataset	F1	W-F1	BOR	Pred/True Boundaries
DialSeg711	0.434	0.767	2.53	5167/2042
SuperSeg	0.609	0.584	0.81	2376/2923
TIAGE	0.368	0.512	0.76	157/207

Table 4: Stage 3 final test results. W-F1 and BOR are consistent with the coherence diagnostics in Table 5.

density, and calibration. Final test results are shown in Table 4.
Calibration. We learned a single global temperature of $T = 0.976$ for this run. In practice, temperature scaling modestly improves score calibration and threshold stability, but it does not eliminate cross-dataset threshold non-transfer: differences in annotation granularity continue to dominate the relationship between score thresholds and boundary density.

8 Segment Coherence Diagnostics

These coherence diagnostics are essential for interpreting boundary-based metrics under granularity mismatch. In particular, they distinguish between two qualitatively different outcomes that exact-match metrics conflate: (1) segmentations that introduce additional but coherent subtopic boundaries, and (2) segmentations that fragment topics into incoherent or mixed segments.

As shown in Table 5, the proposed model maintains high purity across datasets even when boundary density exceeds that of the gold annotation. This indicates that additional predicted boundaries correspond to meaningful conversational structure rather than noise. In contrast, baseline methods with matched boundary density achieve competitive boundary metrics but exhibit lower or unstable coherence, underscoring that density alignment alone does not guarantee usable segmentation.

Taken together, these results demonstrate that segment coherence metrics are necessary to interpret boundary accuracy meaningfully when annotation granularity varies, and that high BOR should not be interpreted as failure in the presence of high purity and coverage.

8.1 Worked Example

We illustrate the effect of annotation granularity on evaluation metrics using a short dialogue example with both coarse gold boundaries and fine-grained model predictions.

This example makes concrete the failure mode identified throughout the paper. The predicted boundaries correspond to coherent conversational subtopics that are omitted by the coarse gold annotation. Under exact-match evaluation, these additional boundaries are treated as false positives, substantially reducing F1 despite correct identification of the major topic transition. The example therefore highlights that the

Dataset	W-F1	BOR	Purity	Coverage
DialSeg711	0.767	2.53	0.962	0.651
SuperSeg	0.584	0.81	0.847	0.915
TIAGE	0.512	0.76	0.785	0.896

Table 5: Model results with segment coherence diagnostics. High purity with elevated BOR indicates coherent fine-grained segmentation that does not match the coarser gold annotation density.

dominant error is one of boundary density rather than boundary placement, and motivates evaluation criteria that distinguish granularity mismatch from detection failure.

9 Discussion

The results demonstrate that boundary scoring generalizes more robustly than boundary selection under heterogeneous annotation granularity. Baseline comparisons in Table 1 expose a central weakness of standard boundary-based metrics: random and periodic strategies achieve competitive W-F1 and purity scores when their output boundary density is matched to the gold annotation, despite lacking any semantic grounding. In contrast, the proposed model yields substantially higher segment purity on datasets with coarse annotations, at the cost of elevated BOR, indicating coherent fine-grained segmentation rather than detection failure.

These findings show that exact-match and tolerance-based boundary metrics conflate segmentation quality with granularity alignment. As illustrated by the baselines, this failure mode is not tied to a particular model architecture but arises whenever boundary density is tuned to match the gold annotation, regardless of the underlying segmentation strategy. Consequently, direct comparison against additional semantic segmentation models would not alter the core evaluation failure identified here.

Additional experiments with classical change-detection and drift-based strategies further support this conclusion. Such methods (e.g., cumulative-sum and time-aware change detection) often achieve higher W-F1 than the neural scorer by aggressively increasing boundary density, frequently yielding boundary oversegmentation ratios well above 1.0. This demonstrates that strong boundary accuracy under tolerant metrics can be driven by recall-dominated strategies rather than by calibrated segmentation aligned with downstream use. The proposed evaluation objective makes this trade-off explicit: methods with higher W-F1 but elevated BOR are correctly identified as oversegmenting, while more conservative scorers exhibit lower boundary accuracy but better density control.

Across the methods examined in this study, three qualitatively distinct segmentation regimes emerge:

- **Conservative strategies** (e.g., SpeechAct, SummaryProbe) undersegment, producing BOR < 1 and low recall.
- **Balanced strategies** (e.g., Neural(fine), commitment-based selection) achieve BOR ≈ 1 with high segment purity, yielding calibrated segmentation aligned with annotation density.
- **Aggressive strategies** (e.g., CUSUM, time-aware drift detection) maximize recall and W-F1 by substantially increasing boundary density (BOR $\gg 1$), resulting in recall-dominated detection rather than calibrated segmentation.

In practice, selection thresholds should therefore be chosen based on intended downstream use rather than optimized solely to match a particular annotation density. Boundary density metrics such as BOR provide a simple diagnostic for tuning selection behavior without retraining the scoring model.

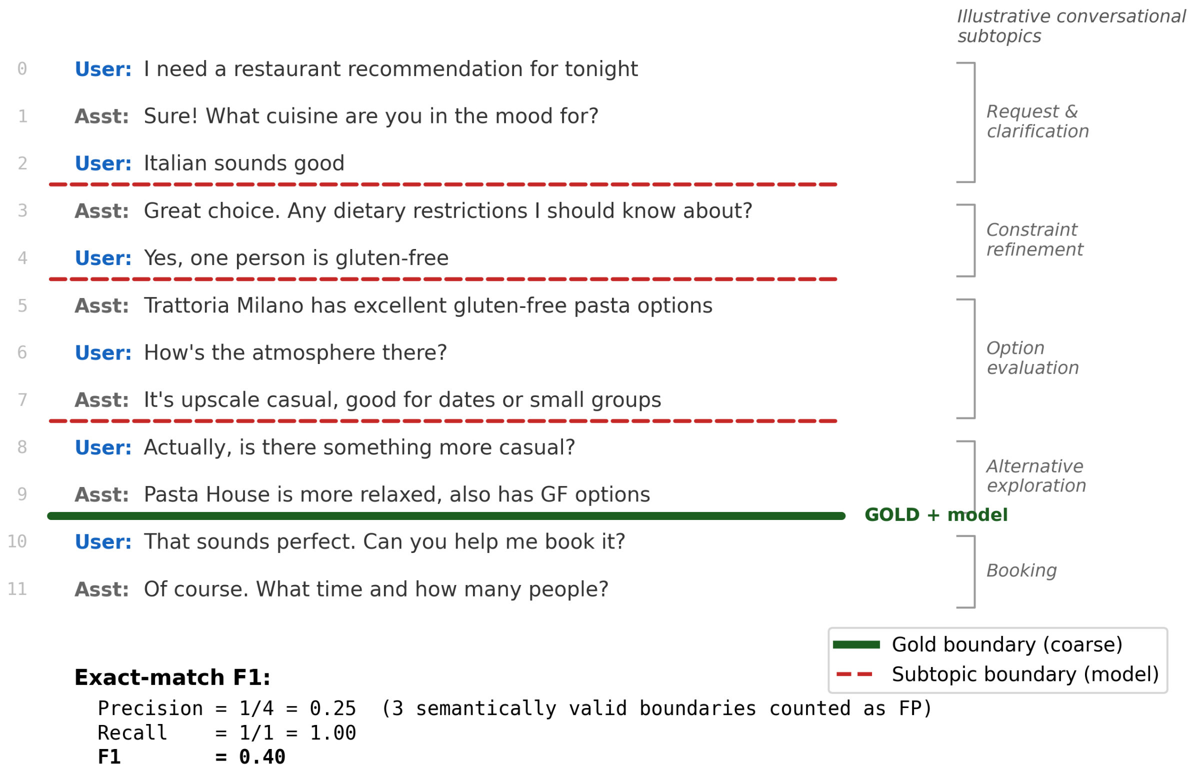

Figure 1: **Granularity mismatch in dialogue topic segmentation.** Gold annotations mark a single coarse topic boundary (browsing → booking), while the model identifies multiple semantically meaningful subtopic transitions. The fine-grained boundaries correspond to coherent conversational phases and are not boundary-positioning errors. Under exact-match evaluation, three valid boundaries are counted as false positives, yielding $\mathrm{F1} = 0.40$ despite correct identification of the coarse transition. The error arises from a mismatch in boundary *density* and segmentation granularity rather than from incorrect boundary detection.

Practical guidance for choosing boundary density. The appropriate boundary selection behavior depends on how segmentation outputs are used downstream. Based on the empirical results in this study, we offer the following high-level guidance:

- **Summarization and context truncation.** When topic boundaries trigger summarization or removal of prior turns, missing a major boundary can be more harmful than introducing extra minor boundaries. Conservative selection is therefore preferable: a lower boundary density (BOR < 1) reduces the risk of prematurely discarding relevant context, even if some distinct topics are merged.
- **Retrieval and navigation.** For applications that rely on topic boundaries to index or retrieve relevant conversation segments, finer-grained segmentation can be beneficial provided segments remain coherent. In this regime, higher boundary density (BOR > 1) is acceptable as long as purity remains high, indicating that additional boundaries correspond to meaningful subtopics rather than noise.
- **Exploratory analysis and annotation.** When segmentation is used for qualitative analysis, annotation support, or exploratory inspection, fine-grained boundaries are often desirable. In such settings, boundary density may substantially exceed that of existing annotations, and evaluation should emphasize segment coherence over exact boundary alignment.

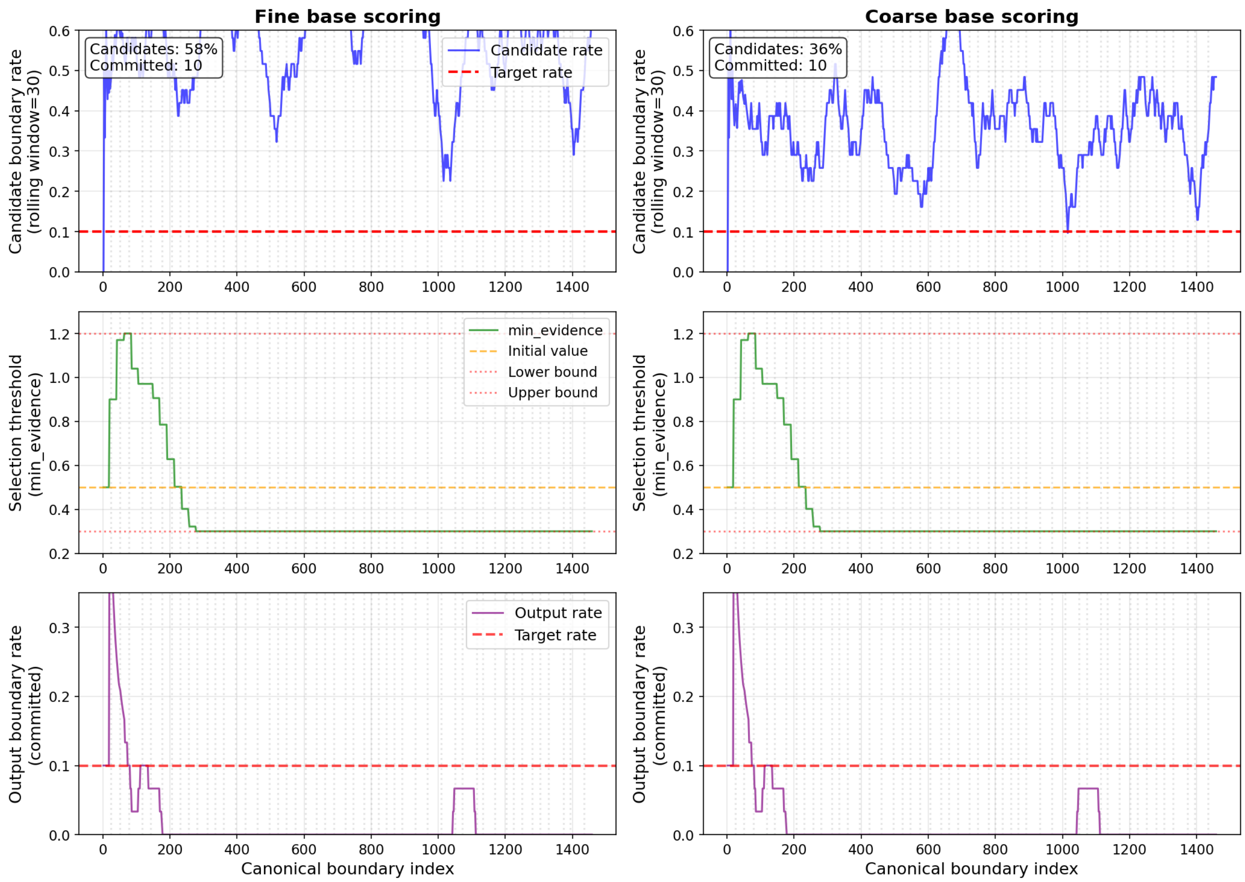

Figure 2: **Adaptive boundary selection normalizes scoring granularity.** Left: fine-grained boundary scoring; right: coarse-grained scoring. *Top row* shows candidate boundary rates, which differ substantially (approximately 58% vs. 36%). *Middle row* shows the adaptive selection threshold (`min_evidence`). *Bottom row* shows the resulting output boundary rate after selection. Despite different scoring behavior, adaptive selection intentionally produces similar output boundary rates; the similarity of the middle and bottom panels is therefore expected and reflects successful decoupling of boundary scoring from boundary selection.

These guidelines reinforce the central claim of this paper: boundary density is not an intrinsic property of a conversation, but a design choice that should be tuned to application requirements rather than optimized to match a given annotation scheme. This dependence on selection behavior motivates treating boundary selection as a distinct, explicit step.

9.1 Boundary Selection as a Separate Step

Separating scoring from selection makes boundary density an explicit control parameter instead of a hidden consequence of threshold tuning.

Figure 2 illustrates why separating scoring from selection matters: different scoring granularities produce different candidate rates, while explicit selection can normalize output boundary counts.

9.2 Computational Cost

Boundary selection is computationally negligible: it operates on a vector of scores and applies thresholding and minimum spacing. Boundary scoring dominates cost because it requires running the neural model over candidate positions (message pairs). The separation therefore has practical value in deployed systems: selection can be retuned per dataset or application without recomputing scores, while recomputing scores requires model inference.

9.3 On Dataset Choice and External Validity

Public dialogue datasets are necessarily imperfect proxies for real human–AI interactions. While several conversational AI datasets have been released, we are not aware of any publicly available corpus of long, open-ended human–AI conversations with turn-level topic boundary annotations suitable for evaluating segmentation granularity. Our claims therefore concern evaluation behavior under heterogeneous annotation schemes rather than direct modeling of user behavior. The consistency of failure modes across diverse datasets indicates that these effects arise from evaluation practice rather than dataset-specific artifacts.

10 Limitations and Future Work

This work does not introduce a new neural backbone for dialogue topic segmentation. Instead, it contributes (i) an evaluation objective that makes boundary density and segment coherence explicit and (ii) a topic segmentation algorithm that separates boundary scoring from boundary selection. We do not include a direct ablation comparing synthetic pretraining against training from scratch. Preliminary experiments indicate that splice-boundary pretraining primarily stabilizes early training dynamics rather than improving final performance, but a systematic ablation remains an important direction for future work.

Purity and coverage are computed with respect to the same gold annotations whose granularity we argue is often mismatched to model behavior. Accordingly, these measures should not be interpreted as independent estimates of semantic coherence. Rather, they function as relative diagnostics: high purity rules out incoherent mixing of gold topics, while elevated BOR distinguishes fine-grained subdivision from boundary noise. Independent coherence measures (e.g., embedding-based similarity or human judgments) would be valuable complements, but are not required for the present goal of diagnosing evaluation failure modes under mismatched annotation granularity.

Statistical variability. We do not report confidence intervals or significance tests for the metrics presented in this work. Our primary claims concern qualitative failure modes and regime-level behavior (e.g., recall-dominated oversegmentation versus calibrated segmentation), which manifest as large and consistent differences in boundary density and coherence across datasets and methods. In preliminary experiments, variance across random seeds and training runs was small relative to these effects and did not alter the qualitative conclusions. Future work could include multi-run statistical reporting for finer-grained comparisons within a given segmentation regime, but such analysis is not required to support the central evaluation claims made here.

Inter-annotator agreement (IAA) statistics could also provide additional empirical support for the claim that topic boundaries are granularity-dependent rather than objective. Several dialogue segmentation datasets report IAA in their original publications (e.g., DialSeg711 [9], SuperDialseg [8]), often showing only moderate agreement even under controlled annotation protocols. However, these statistics are reported using heterogeneous definitions and units (e.g., boundary placement, segment overlap, or task-level agreement), making direct comparison across datasets difficult. Because the present work focuses on evaluation behavior under fixed annotation schemes rather than on the reliability of those schemes themselves, we do not re-analyze annotator agreement here. Integrating IAA into a unified analysis of granularity effects remains an important direction for future work.

Additional directions include evaluating published dialogue segmentation methods under the proposed evaluation objective to confirm that the observed granularity effects are method-agnostic, and exploring alternative boundary selection rules that adapt boundary density to application constraints rather than annotation conventions. Further study is also needed to relate boundary density preferences to specific downstream uses such as summarization, retrieval, or conversational memory. Finally, future work could replace discrete boundary representations with continuous or multi-scale topic representations (e.g., topic clouds) that better capture overlapping, gradual, or revisited conversational structure.

11 Conclusion

Dialogue topic segmentation does not admit a single ground truth. Exact boundary matching conflates detection correctness with segmentation granularity and can obscure coherent segmentation behavior under heterogeneous annotation schemes. By separating boundary scoring from boundary selection and evaluating boundary density and segment coherence alongside tolerant boundary accuracy, this work provides a principled basis for diagnosing evaluation failures across datasets.

A central empirical finding is that boundary-based metrics can be dominated by boundary density alignment: even simple non-semantic baselines achieve competitive F1 and W-F1 scores when their output boundary rate matches the gold annotation. This shows that boundary accuracy alone is insufficient when annotation granularity varies.

The proposed separation is intentionally simple. Its value lies in revealing systematic evaluation failures that arise when granularity is ignored. Our goal is not to claim model superiority, but to motivate reporting practices that make boundary density and segment coherence explicit.

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